Dialogue systems ~~Conversational agents~~ for language learning: state of the art and avenues for research on task-based agents

Serge **Bibauw** [1,2,3] ▪ Thomas **François** [2] ▪ Piet **Desmet** [1]

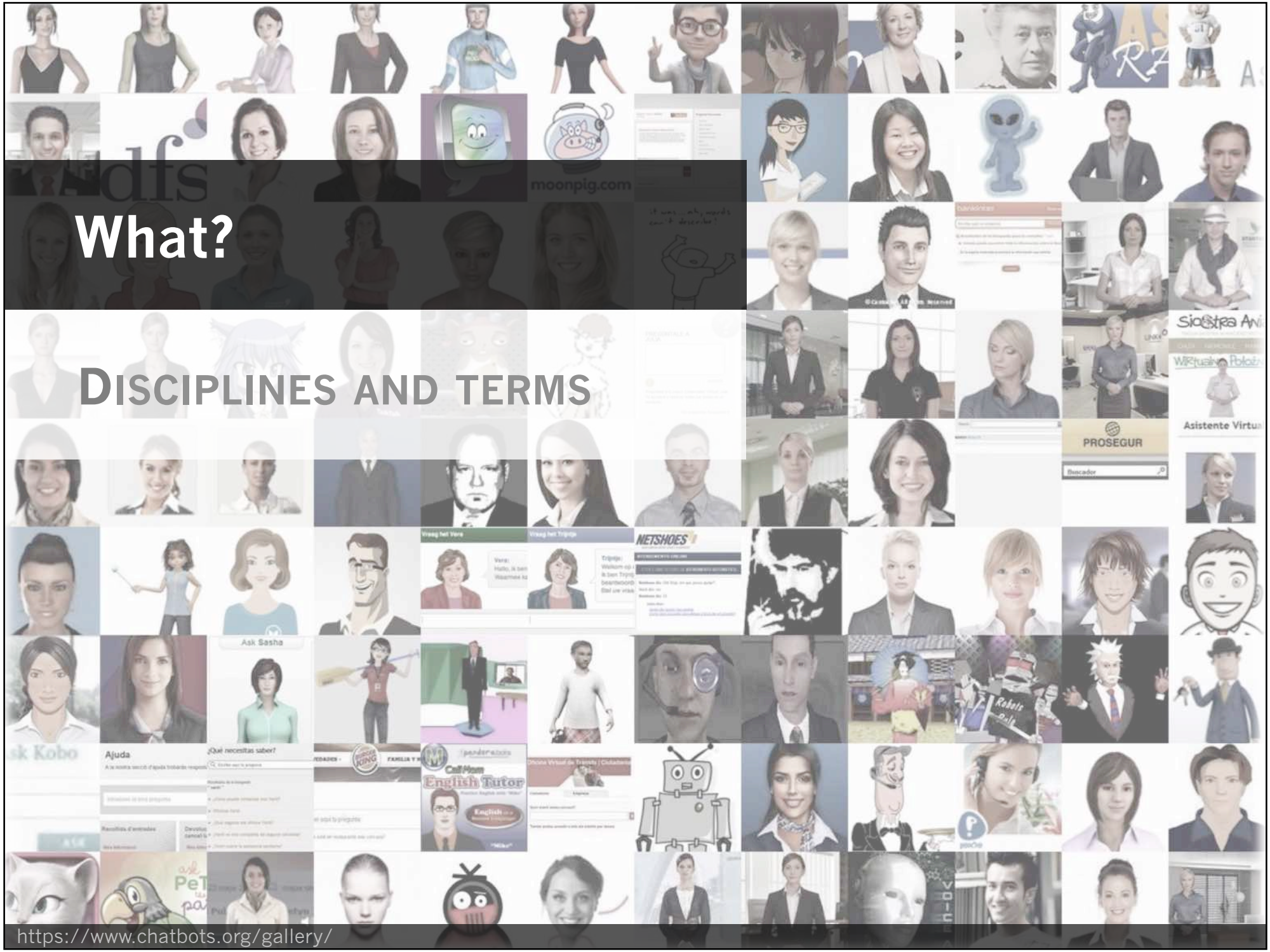
[1] KU Leuven Kulak ▪ iMinds ▪ ITEC
[2] Université catholique de Louvain ▪ CENTAL
[3] Universidad Central del Ecuador

Dialogue systems for language learning

- What? — Disciplines and terms
  - Dialogue systems — Definition and components
- Why? — Rationale
  - What do we know? — Effectiveness studies
- How? — Technological process

What?

Disciplines and terms



https://www.chatbots.org/gallery/

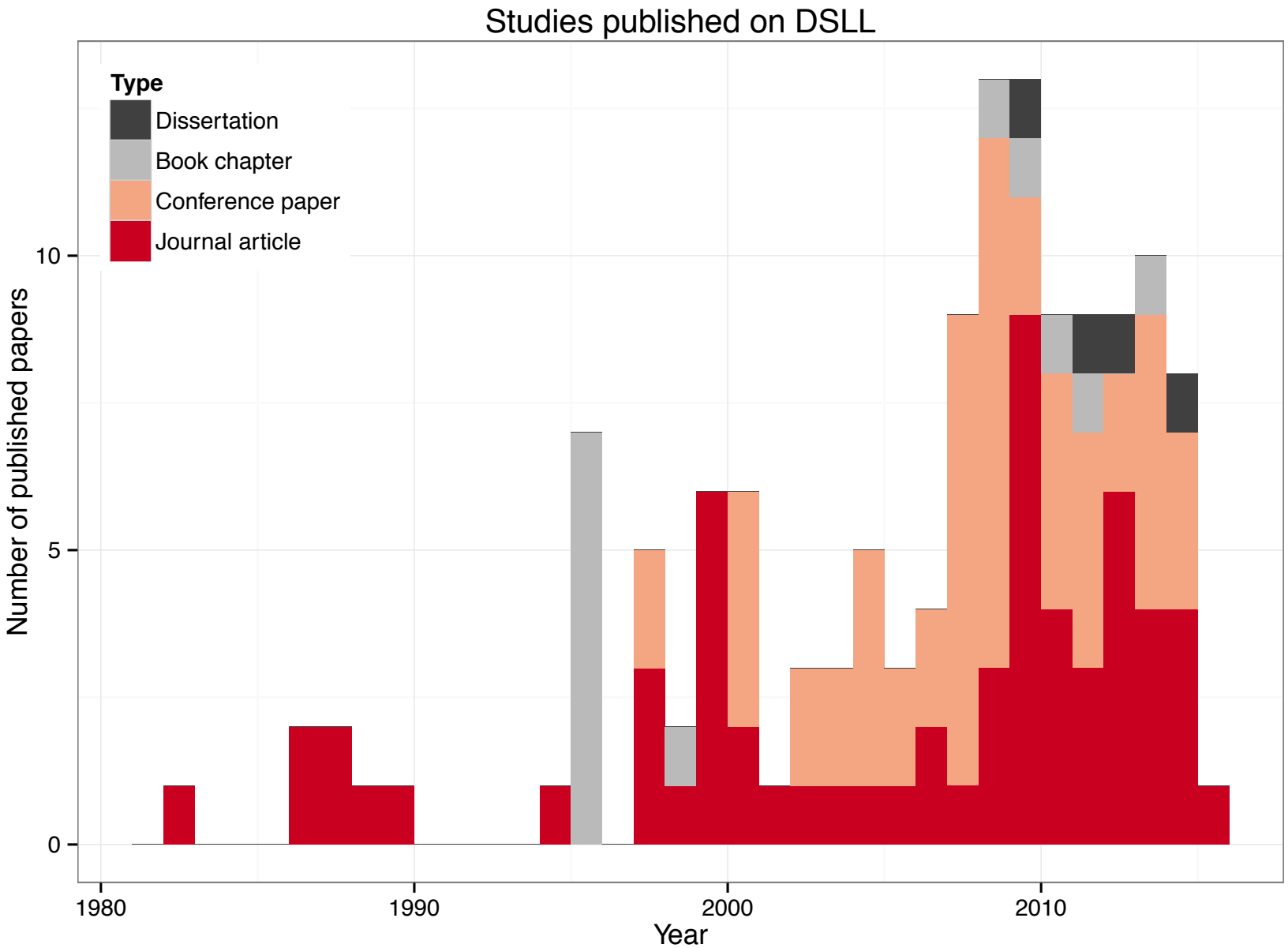
Dialogue system?

- Chatbot
- Conversational agent
- **Dialogue system**
- Spoken dialogue system
- Dialogue-based CALL
- Dialogue simulation
- Dialogue game
- Simulated dialogue
- Speech-interactive CALL
- Conversational language tutoring system
- Conversation simulator
- Communication-based language learning
- Chatterbot
- Conversational companion
- Conversational system
- Conversational ICALL system
- Dialogue-based conversation tutoring
- Dialogue program
- Interactive pedagogical drama
- Microworld interaction
- Virtual agent
- Virtual human
- Pedagogical agent
- Voice-interactive CALL...

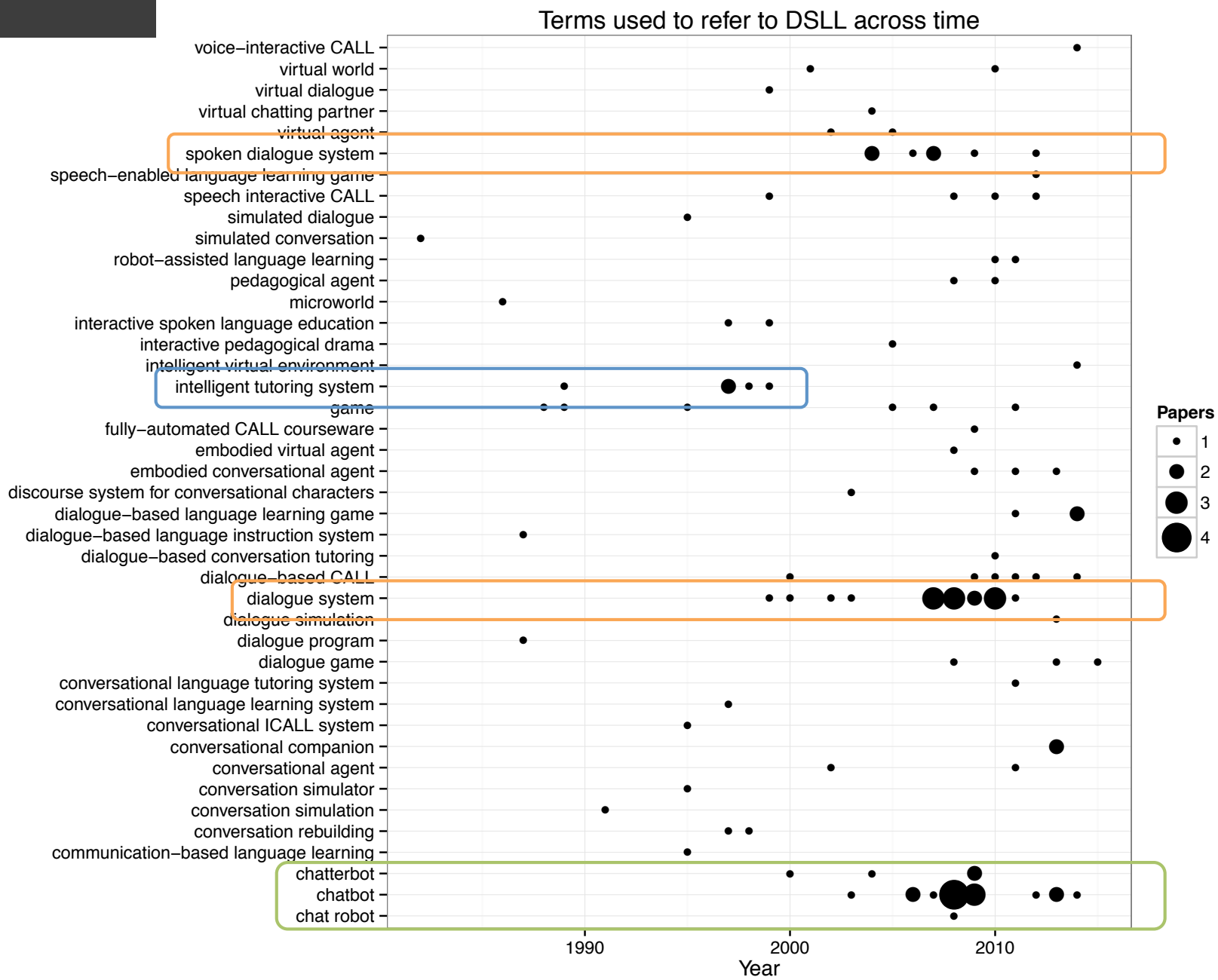
Systematical literature study: Corpus collection

- Systematical search on
  Web of Science, Scopus and ProQuest
  [results: 604 / 494 / 1003 hits]
- Ancestry (citing) and forward (cited in) search
- Final corpus:
  135 published and peer-reviewed papers
  from 1982 to 2015 (April)

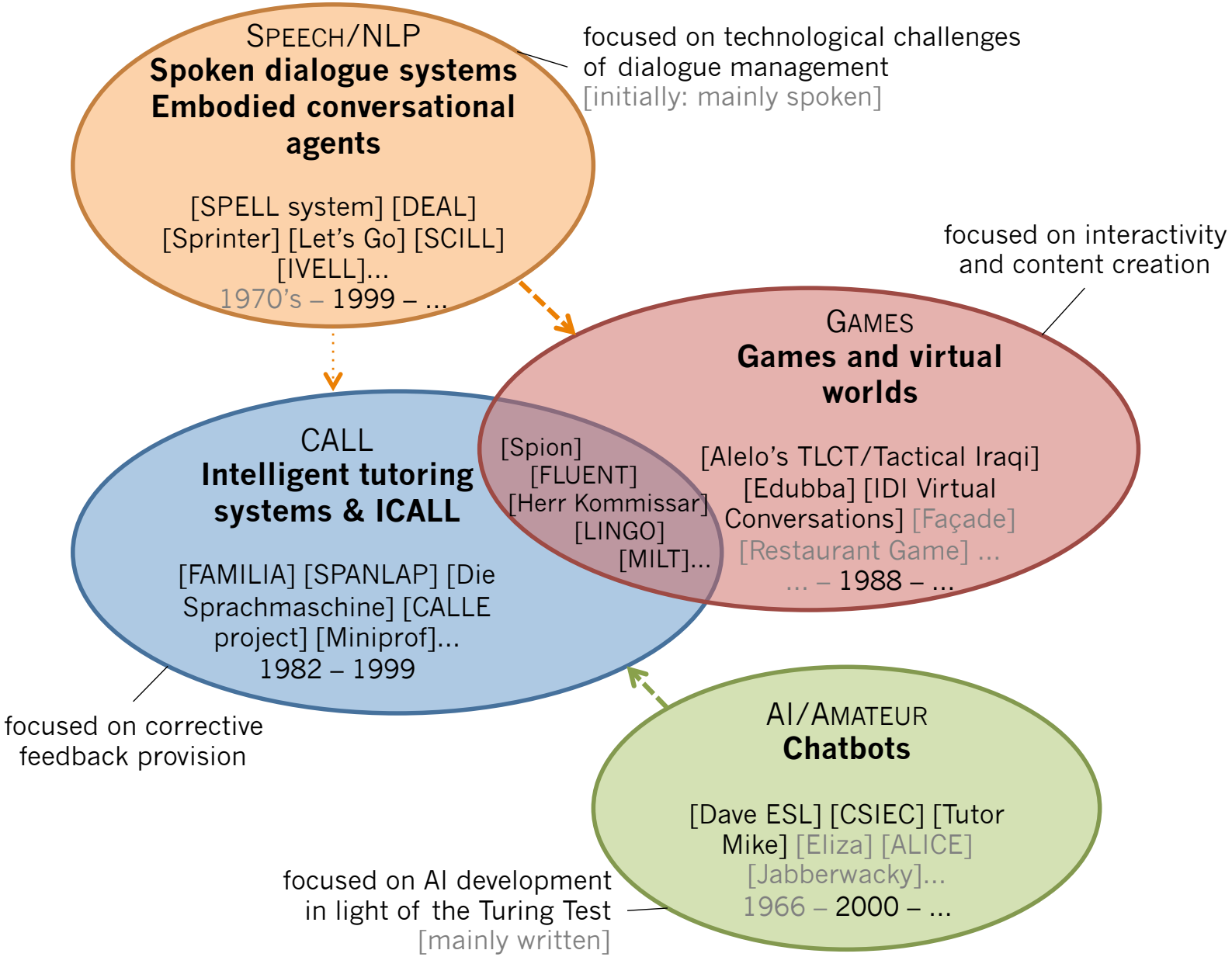
Corpus of studies



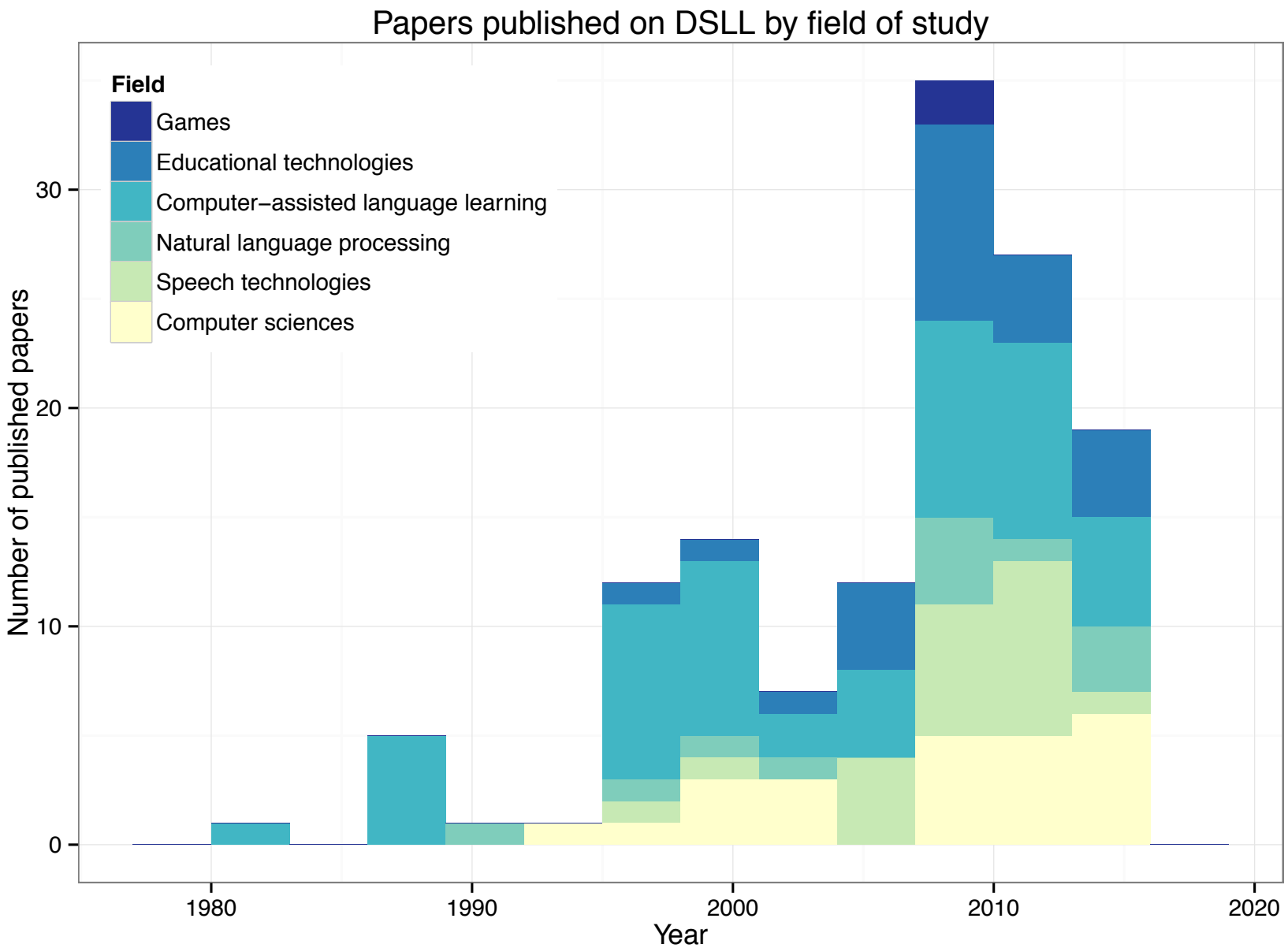
Terms


Terms used to refer to DSLL across time
voice–interactive CALL
virtual world
virtual dialogue
virtual chatting partner
virtual agent
spoken dialogue system
speech–enabled language learning game
speech interactive CALL
simulated dialogue
simulated conversation
robot–assisted language learning
pedagogical agent
microworld
interactive spoken language education
interactive pedagogical drama
intelligent virtual environment
intelligent tutoring system
game
fully–automated CALL courseware
embodied virtual agent
embodied conversational agent
discourse system for conversational characters
dialogue–based language learning game
dialogue–based language instruction system
dialogue–based conversation tutoring
dialogue–based CALL
dialogue system
dialogue simulation
dialogue program
dialogue game
conversational language tutoring system
conversational language learning system
conversational ICALL system
conversational companion
conversational agent
conversation simulator
conversation simulation
conversation rebuilding
communication–based language learning
chatterbot
chatbot
chat robot
1990
2000
2010
Year
Papers
1
2
3
4

Disciplinary approaches

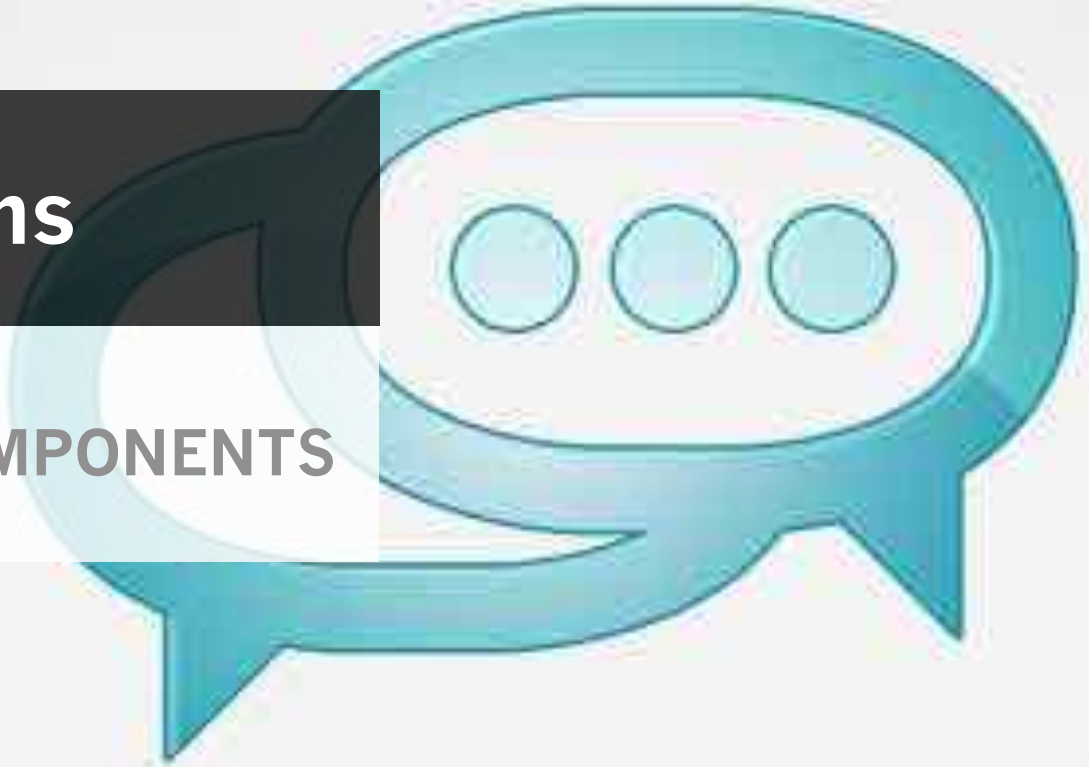


Disciplinary approaches



Dialogue systems

Definition and components



Dialogue System
for unity

http://d2ujflorbtfzji.cloudfront.net/key-image/798a5cac-c78c-479f-a58f-49ab25a920e5.jpg

Different systems?

- Spoken / Written / Multimodal
  e.g. spoken dialogue systems vs. chatbots
- Technologically simple / complex
  e.g. chatbots vs. dialogue systems (Klüwer, 2011)
- Focus on form / Focus on meaning
  e.g. tutorial CALL vs. “communication-based CALL” (Murray, 1995)

Or a single endeavor?

- **Dialogue activity**
  - conversation, talk, information exchange, questions-answers, requests-grants, etc.
- with an **automated agent**
  - as interlocutor, as tutor, as character…
  - virtual, embodied, robot, etc.

DSLL: Towards an operational definition

- **Dialogue** activity
  - conversation, talk, information exchange, questions-answers, requests-grants, etc.
- with an **automated agent**
  - as interlocutor, as tutor, as character…
  - virtual, embodied, robot, etc.

➢ **Dialogue system for language learning** (**DSLL**)

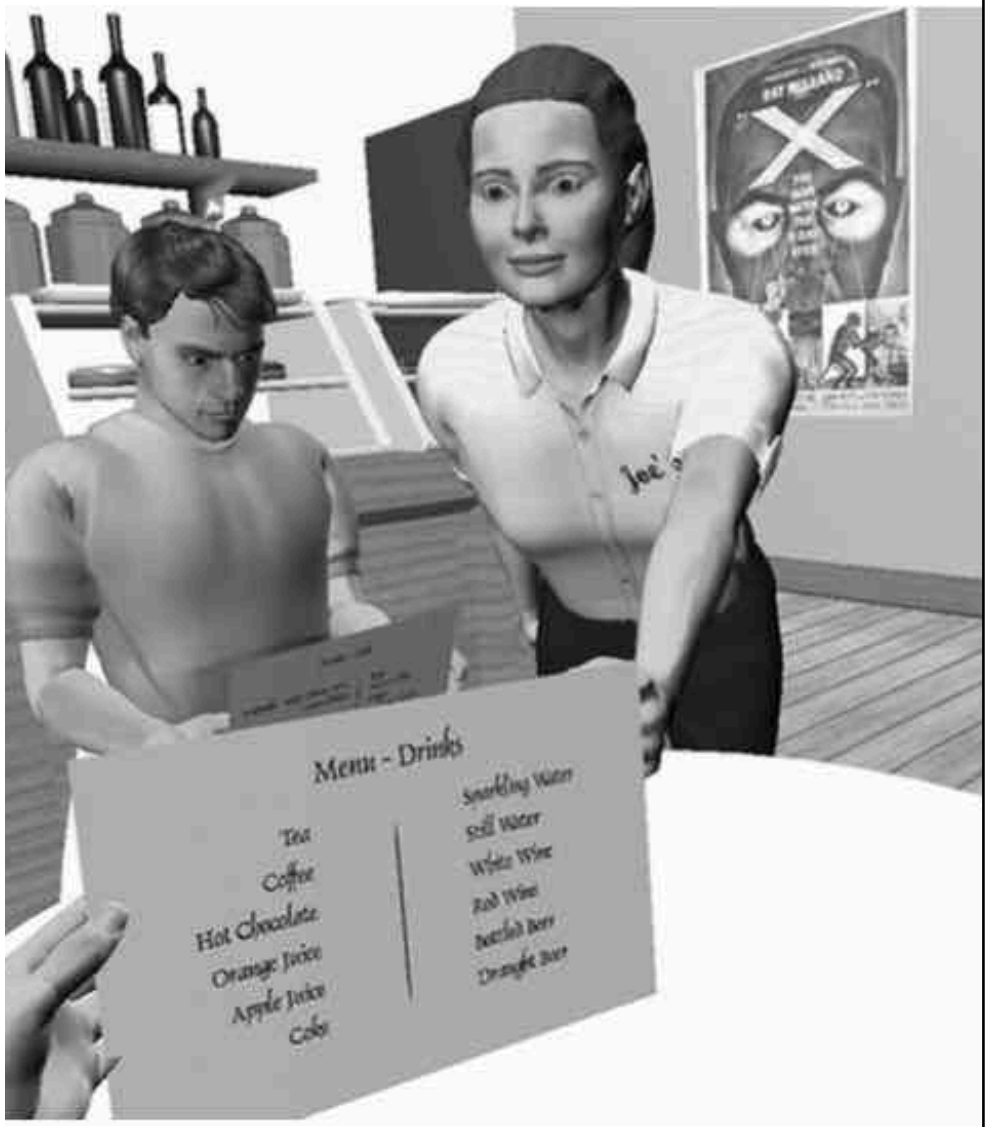
Tutor Mike (Lee)

Practice English with Mike

Mike is very smart. You can ask him any questions. When you want to practice the conversations of the topic you are
www.eslfast.com/robot/, type in "Let's practice topic ..." or "I'd like to practice topic ..." You must type the exact nam
Let's practice topic Weather (2). When you are acting as **A** and Robot Mike is **B**, you must type in the exact sentenc
conversation, including the punctuation. If you make a mistake, it is not controlled practice of the conversation anym
you need to type "Let's practice topic ..." again to get the robot back on track.

http://www.eslfast.com/robot/english_tutor.htm

SPELL system (Morton, Gunson & Jack)



Morton & Jack, 2005, 2010; Morton, Davidson & Jack, 2008; Morton, Gunson & Jack, 2012

Distinctive features

Interlocutor	Human	Human + Automated tutor	Automated interlocutor
Interaction	Monologue	“Mono-initiative” dialogue	Dialogue
Input	Constrained	Partially constrained	Unconstrained
Interactivity	Fixed	Fixed with illusion of choice	Interactive

Interaction ↓	Input ↓	Human	Human + Automated tutor	Automated	Automated	Automated	← Interlocutor
		Interactive	Interactive	Interactive	Fixed with illusion of choice	Fixed	← Interactivity*
Monologue	Constrained input	Registration form	?	?	?	Word(s) completion (*gap-filling*), selection, ordering, etc.	
Monologue	Partially constrained input	?	?	?	Choose from a list of words (various possible sentences)	Choose from a list of words	
Monologue	Unconstrained input	Webinar. Usual web publication	Computer-assisted writing system	?	?	Short answer (with automatic correction)	
Mono-initiative dialogue	Partially constrained input	?	?	**Text adventure game** and microworlds [FLAG, Spion, LINGO, MILT written]	?	Reading aloud prompted utterances (de Wet *et al* 2009)	
Mono-initiative dialogue	Unconstrained input	Questions & answers website [Yahoo Answers]	?	**User-only initiative in chatbot** [CSIEC...]	**Questions-asking system** where answer is not taken into account [ALICE-chan, Saybot]	**Tutorial dialogue**	**Interactive uncons-trained DS**
Dialogue	Unconstrained input	**Computer-mediated communication**: chat, email, videoconference, forum...	**Computer-supported collaborative learning** and **Tutored CMC** [MentorChat, Neckle, C4/Mondo]	Interactive unconstrained **DIALOGUE SYSTEM** [SPELL, Subarashii, DEAL, CSIEC, TLCTS, Façade...]	Free input, but ignored by system [Majestic]	**Embedded dialogue** (Cornillie *et al* 2013)	
Dialogue	Partially constrained input	?	?	?	Choose from a list of words in a dialogue (various correct sentences possible)	Choose from a list of words in a dialogue	**Partially interactive and uncons-trained DS**
Dialogue	Constrained input	?	?	Utterance selection dialogue with multiple conversational paths [Let's Chat, Animated pedagogical agent]	Utterance selection with various correct options, but same reaction [Trace Effect]	Reading aloud prompted utterances in dialogue [Virtual Language Patient, FASOP]	
Mono-initiative dialogue	Constrained input	Request more information form	?	Microworld interaction with utterance selection [VILTS, MILT spoken]	Tutorial dialogue asking for words [ARTUR]	Ordering words (*drag & drop*) or *gap-filling* in a dialogue	All DS

Focus-on-meaning
Successful communication is the main objective

Focus-on-form
Corrective feedback is the main objective

Interaction ↓	Input ↓	Human	Human + Automated tutor	Automated	Automated	Automated	← Interlocutor
		Interactive	Interactive	Interactive	Fixed with illusion of choice	Fixed	← Interactivity
Monologue	Constrained input	9				14	
Monologue	Partially constrained input						
Monologue	Unconstrained input			1			
Mono-initiative dialogue	Partially constrained input			**Text adventure games** [Spion, LINGO, MILT...] 7			
Mono-initiative dialogue	Unconstrained input	1		**Mono-initiative chatbots** [Dave, Mike, CSIEC] 9	**Questions-asking systems** [ALICE-chan, Saybot] 1		
Dialogue	Unconstrained input	2	Computer-supported collaborative learning/ **Tutored SCMC** 5 [MentorChat, Neckle...]	**Interactive & unconstrained DSLL** 82	2	1	
Dialogue	Partially constrained input				1		
Dialogue	Constrained input			3	2	**Reading aloud** prompted utterances [FASOP] 2	
Mono-initiative dialogue	Constrained input			**Utterance selection** [Let's Chat...] 5		**Gap-filling** 1	

Interactive unconstrained DS

Partially interactive and unconstrained DS

All DS

Focus-on-meaning
Successful communication is the main objective

Focus-on-form
Corrective feedback is the main objective

Why?

Rationale for DSLL

http://media.timeout.com/images/resizeBestFit/100238249/660/370/image.jpg

SCMC (chat) ⇒ L2 development

- SCMC ⇒ L2 oral proficiency
  (Payne & Whitney, 2002; Payne & Ross, 2005;... Ziegler, 2013; Lin, 2015)
- SCMC ⇒ L2 oral proficiency, more than face-to-face
  (Sykes, 2005; Lin, 2015; although not confirmed by Ziegler, 2013)
- Why?
  - **attention** to form (**noticing** and **feedback**)
  - lower anxiety levels (Satar & Özdener, 2008)

DSLL ⇢ L2 development

- In foreign language teaching contexts: interactions in L2 often very rare

⇒ "Virtual immersion" (Ellis & Bogart, 2007; Fryer & Carpenter, 2006)

- Autonomous practice in a communicative, cultural, authentic and interactive task
  (Wachowicz & Scott, 1999; Fryer & Carpenter, 2006)

DSLL ⇢ L2 development: advantages over SCMC

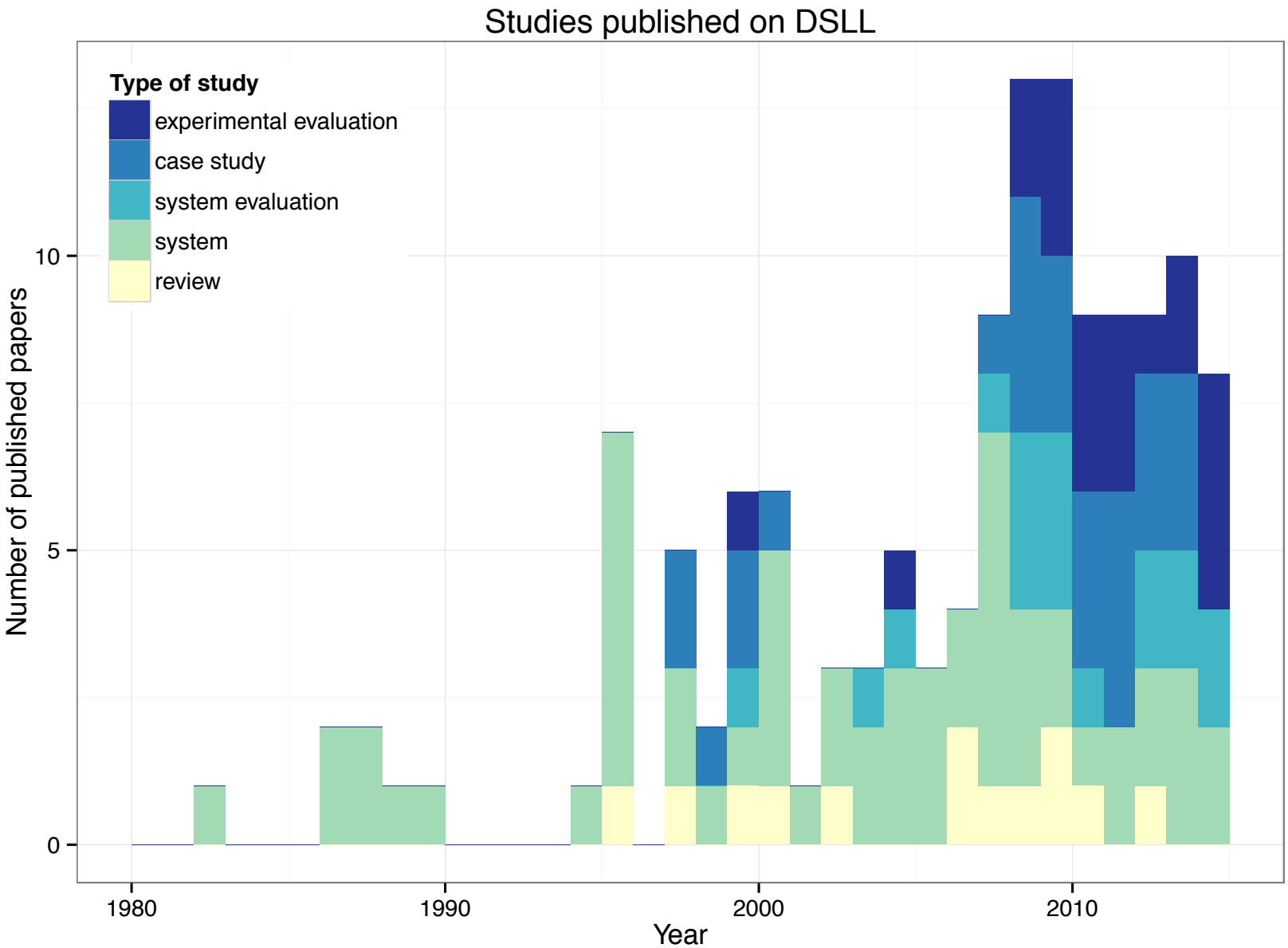
- Available at any moment
- Learner can go at his own pace (repeat, etc.)
- Low anxiety environment
- Potentially fully controllable learning environment
  (e.g. feedback, learner modeling and adaptivity, motivational support)

What do we know?

Effectiveness studies

https://databustersblog.files.wordpress.com/2014/10/control-groups.jpg

Types of **studies** on DSLL

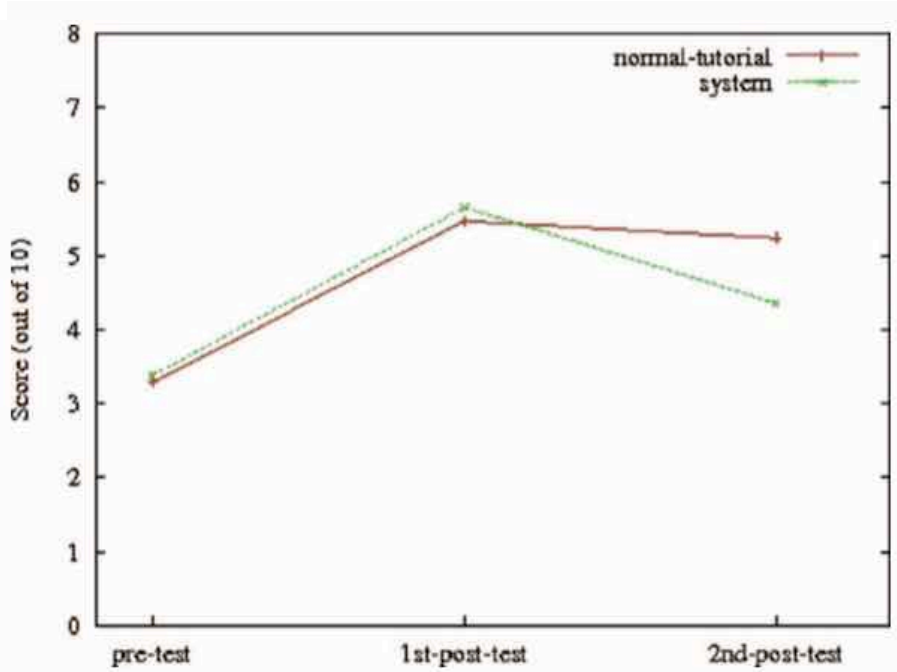


Effectiveness studies

Number of studies by dependent variables and methodological design	**Perceived usefulness**	**Impact on motivation and attitude**	**Impact on language knowledge and skills**
Post	14	1	3
Pre-post		7	22
Pre-post + control		4	12
Pre-post + control with significant results		0	2

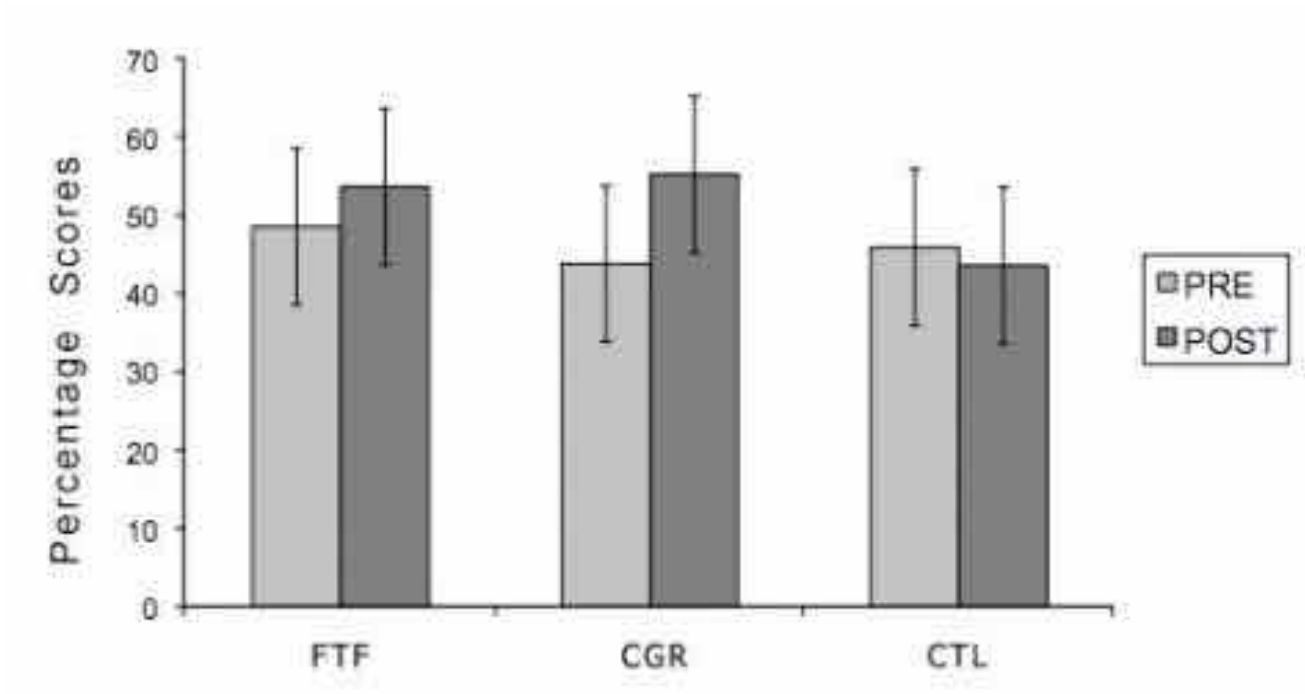
Experimental studies on DSLL's impact on learning

- Vlugter, Knotta, McDonald & Hall (2009)
  [Te Kaitito]: written DSLL tutorial in Māori
  - compares DSLL tutorial with in-class tutorial about Māori personal pronoun system



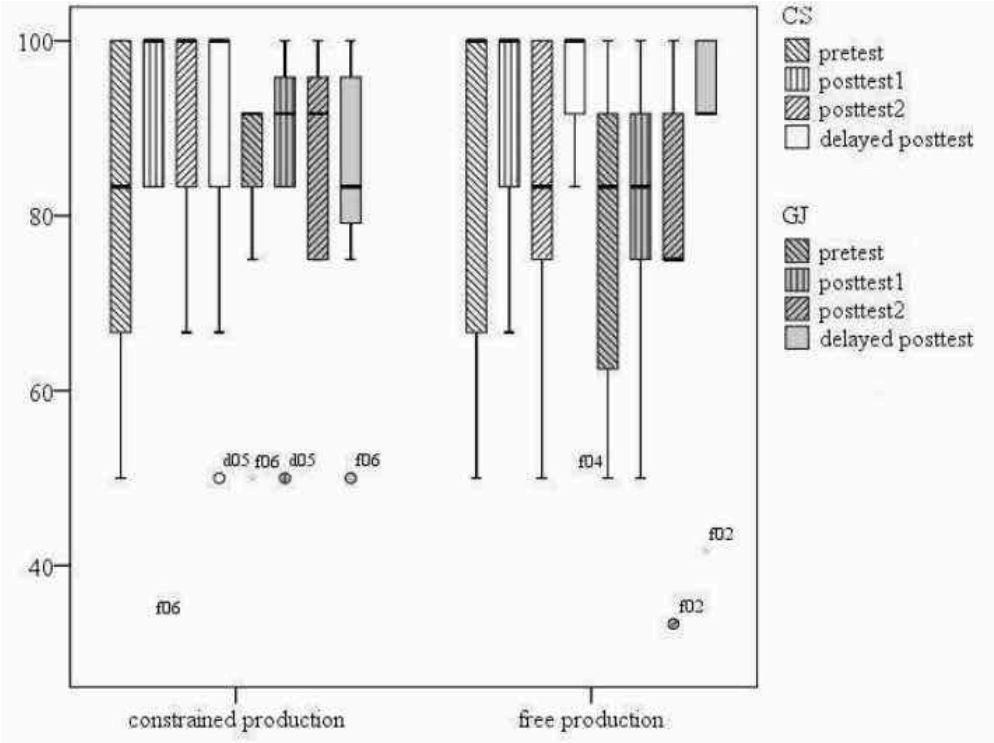
Experimental studies on DSLL's impact on learning

- Petersen (2010)
  [Sasha]: written DSLL in English providing corrective feedback (recast) and answering questions in a "spot the differences" task
  - compares Sasha vs. face-to-face interaction
    => impact on question construction



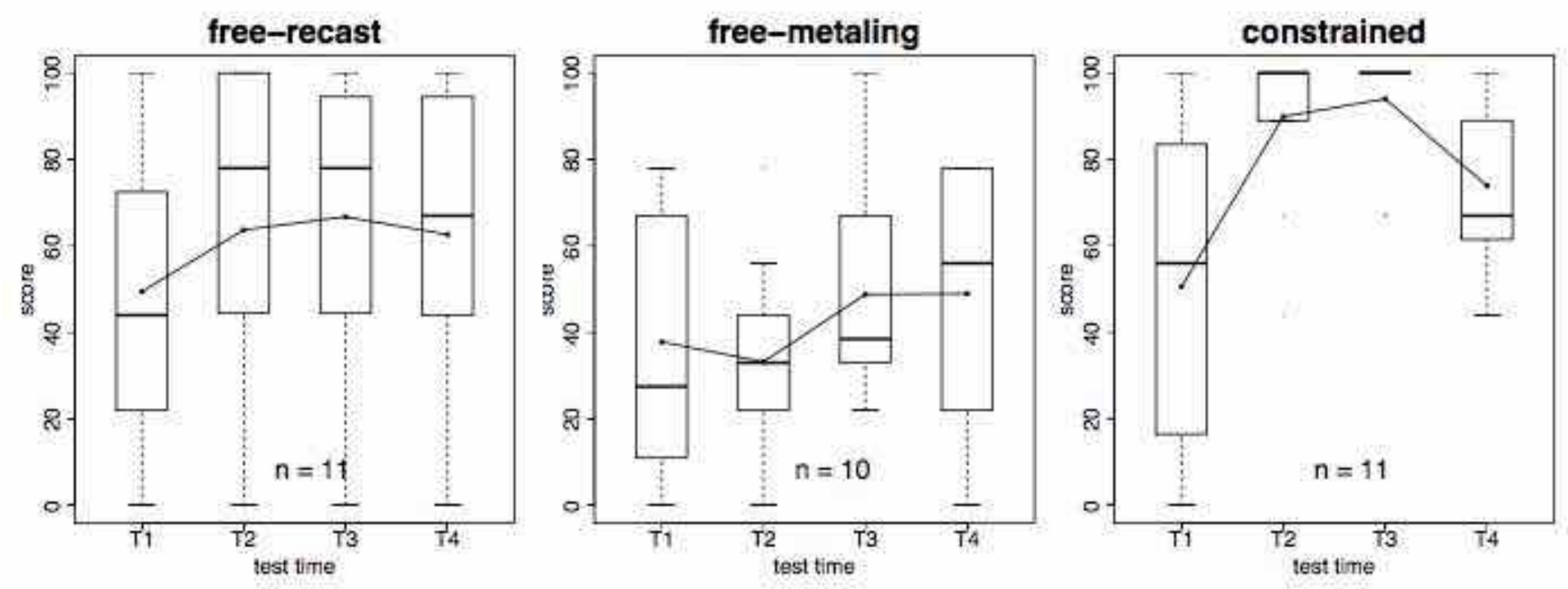
Experimental studies on DSLL's impact on learning

- Wolska & Wilske (2010a)
  Written DSLL in German
  - compare **free production** vs. **constrained production** (*gap filling*) => impact on sentence construction and grammaticality judgement
  - N=15
    n=7



Experimental studies on DSLL's impact on learning

- Wilske & Wolska (2011)
  - compares free production with either recasts or metalinguistic feedback, and constrained input
    => impact on sentence construction & grammaticality judgement
  - N=30 / n=9

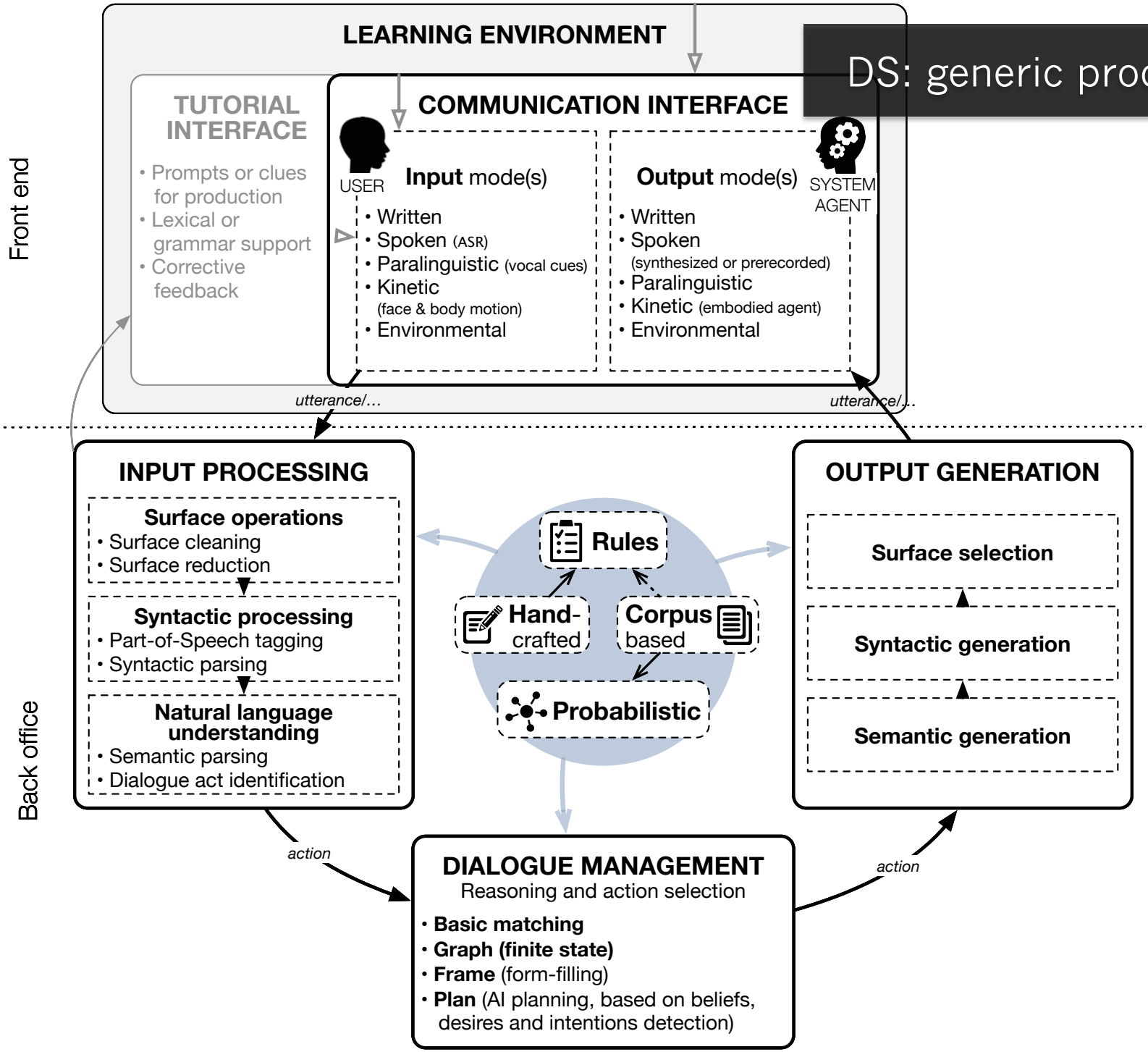


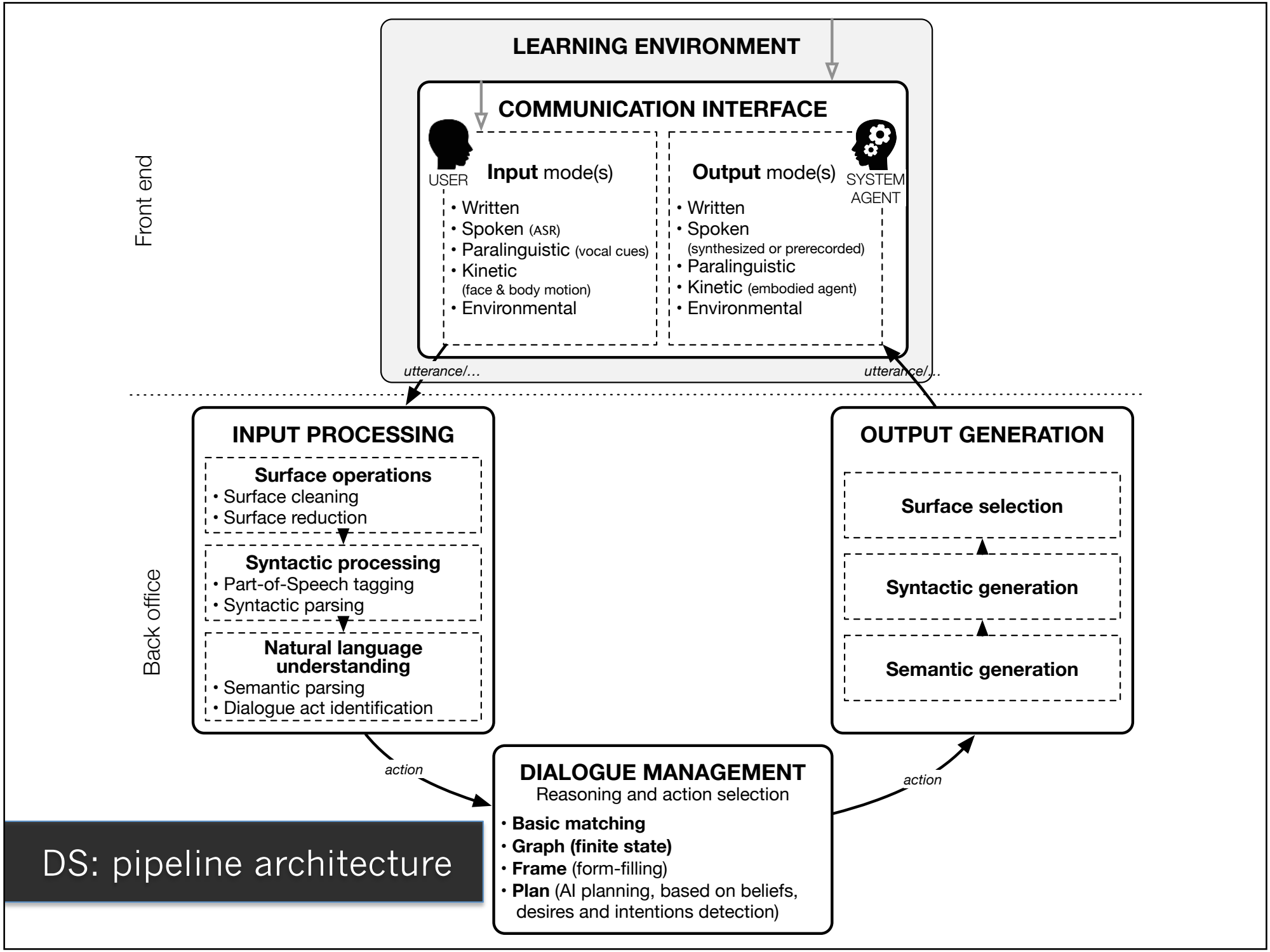
How?

Technological process

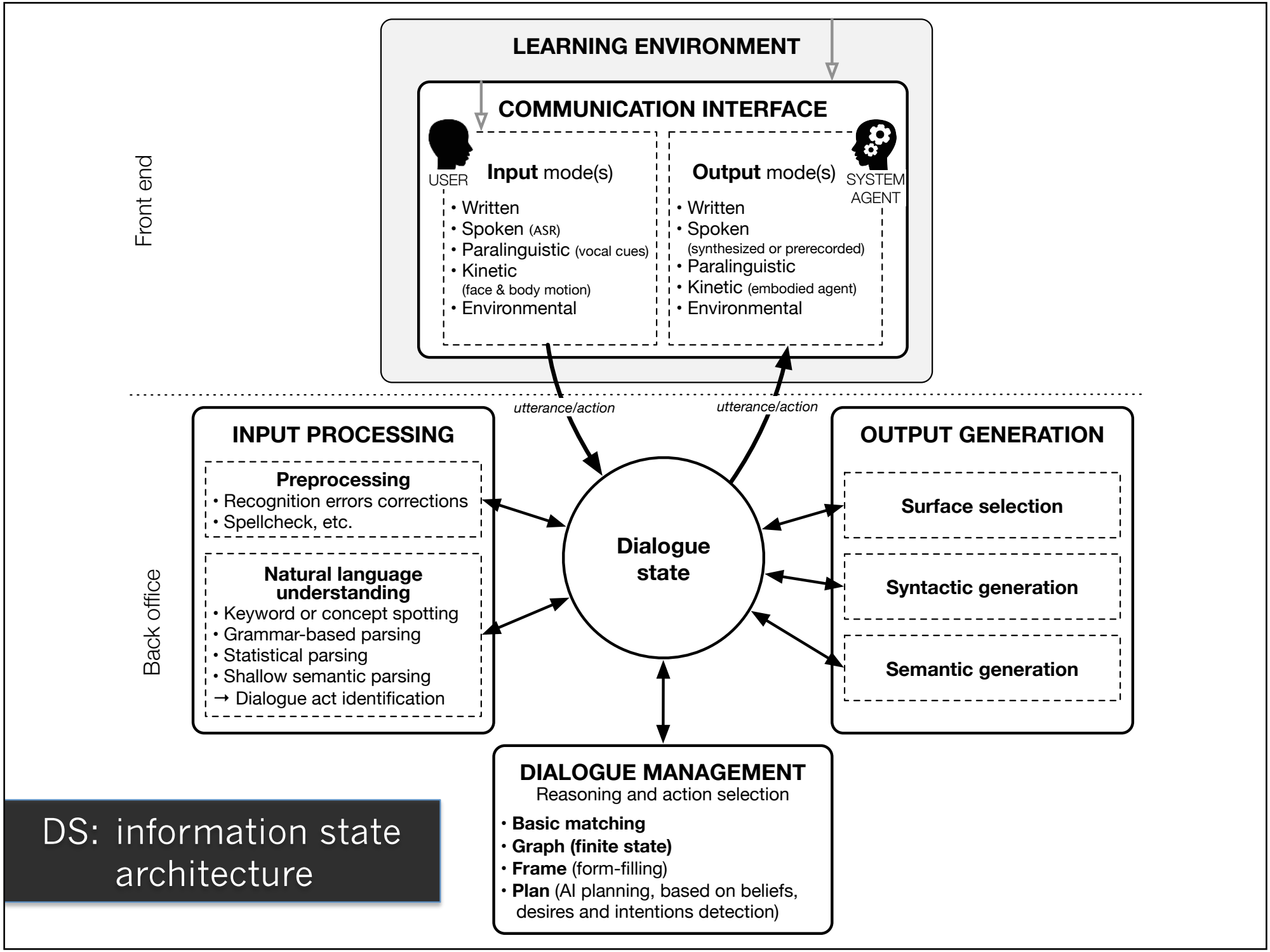
```
<li>Doesn't everyone have a </li>
</random><star/>.<br />
I have
<random>
<li>A lot of friends on the Internet.</li>
<li>A great programmer.</li>
<li>A whole lot of original answers.</li>
<li>A plan for a robot body.</li>
</random>
</template>
</category>

<category>
<pattern>DO YOU KNOW *</pattern>
<template>
<think><set name="it"><star/></set></think>
<random>
<li>I don't know if I know <star/>. Where would I have heard about it?</li>
<li>I know very little about <star/>.  Tell me more.</li>
<li>No, tell me about him, her or it.</li>
</random>
</template>
</category>

<category>
<pattern>DO YOU KNOW WHAT * IS</pattern>
<template>
<srai>WHAT IS <star/></srai>
</template>
</category>

<category>
<pattern>DO YOU LIKE *</pattern>
<template>
<think><set name="it"><star/></set></think>
<random>
<li>I don't know if I like <star/>.</li>
<li>Some people like it.</li>
<li>Do you like it?</li>
<li>How old are you?</li>
<li>I've heard other people say they like that.</li>
</random>
<random>
<li>I like pets.</li>
<li>But I like cats.</li>
<li>Though I like cats and dogs.</li>
<li>I enjoy working with people.</li>
<li>I have a stimulating relationship with <bot name="master"/>.</li>
<li>I love horses.</li>
<li>Someone said they like guns.</li>
```

DS: generic process



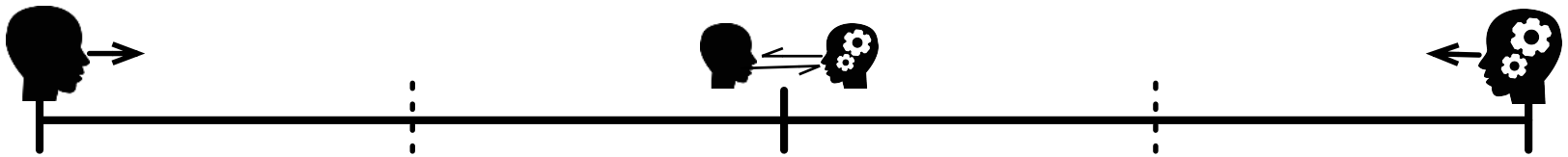


DS: pipeline architecture



DS: information state architecture

Initiative management in dialogue systems



user-only initiative

- question-answering systems
- text adventure games & microworld interactions

Main problems:
- References processing
- Information retrieval

default user initiative

- most chatbots

mixed initiative

- cooperative dialogue
- most task-oriented dialogue systems

Main problems:
- Natural language understanding
- Dialogue modeling and dialogue management
- Natural language generation

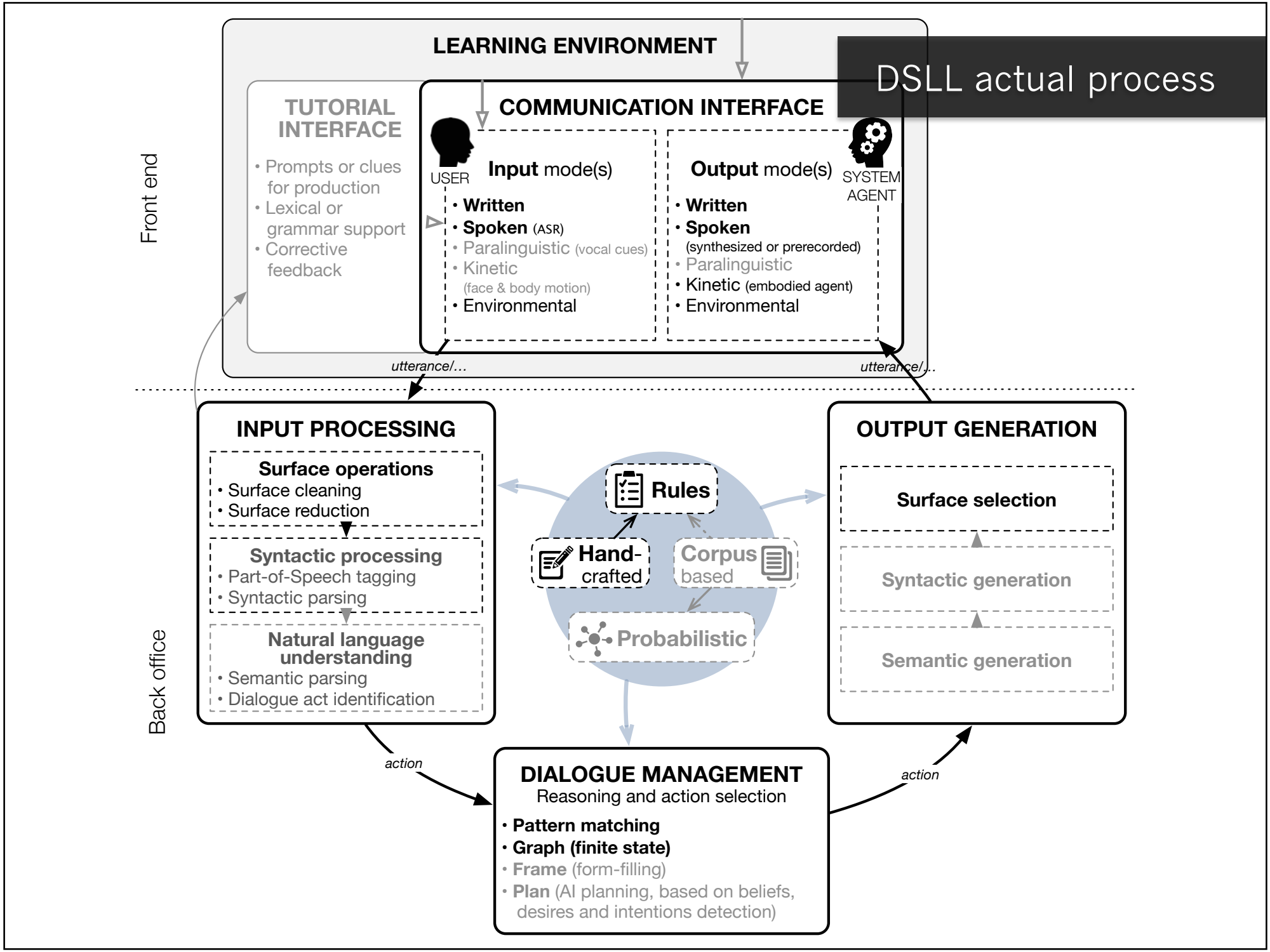
default agent initiative

- tutorial dialogue (most ITS)

system-only initiative

- question-asking systems
- rigid tutorial dialogue

Main problems:
- Materials scripting
- Corrective feedback generation


DSLL actual process
LEARNING ENVIRONMENT
TUTORIAL INTERFACE
• Prompts or clues for production
• Lexical or grammar support
• Corrective feedback
COMMUNICATION INTERFACE
USER
Input mode(s)
• Written
• Spoken (ASR)
• Paralinguistic (vocal cues)
• Kinetic (face & body motion)
• Environmental
Output mode(s)
SYSTEM AGENT
• Written
• Spoken (synthesized or prerecorded)
• Paralinguistic
• Kinetic (embodied agent)
• Environmental
Front end
utterance/...
utterance/...
INPUT PROCESSING
Surface operations
• Surface cleaning
• Surface reduction
Syntactic processing
• Part-of-Speech tagging
• Syntactic parsing
Natural language understanding
• Semantic parsing
• Dialogue act identification
Back office
Rules
Hand-crafted
Corpus based
Probabilistic
OUTPUT GENERATION
Surface selection
Syntactic generation
Semantic generation
action
action
DIALOGUE MANAGEMENT
Reasoning and action selection
• Pattern matching
• Graph (finite state)
• Frame (form-filling)
• Plan (AI planning, based on beliefs, desires and intentions detection)

Conclusions

- **What?**
  - Operational definition of DSLL based on systematic literature study:
    - "Dialogue activity with an automated agent, ideally with unconstrained input and interactive dialogue sequence"
- **Why?**
  - Insufficient experimental results to demonstrate effectiveness for language language

  ⇒ Need for more effectiveness research
- **How?**
  - Advancements in dialogue systems haven't yet been applied to DSLL:
    - natural language understanding
    - information state-based
    - data-driven / probabilistic models

  ⇒ Need for more technological research

Thank you!

Dialogue systems for language learning:
state of the art and avenues for research
on task-based agents

Serge **Bibauw**
serge.bibauw@kuleuven.be

Thomas **François**
thomas.francois@uclouvain.be

Piet **Desmet**
piet.desmet@kuleuven.be

KU LEUVEN iMinds itec UCL Université catholique de Louvain CENTAL centre de traitement automatique du langage UNIVERSIDAD CENTRAL DEL ECUADOR